Inventory Stock Level Audit

Comparing system records to physical counts, classifying SKUs, and finding stockouts and overstock

Scope	150 SKUs across 6 categories (155 raw rows)
Method	System vs physical count, ABC analysis, reorder-point check, turnover
Tools	Python (pandas, matplotlib), Excel (formulas, pivot tables)
Data	Synthetic, generated for learning purposes

Headline results

Discrepancy rate (SKUs not matching)	28.7%
Exact-match record accuracy	71.3%
Net variance (physical minus system)	-$45,248
Inventory value: system vs physical	$905,596 vs $860,348
Inventory turnover	4.8x per year (76 days of stock)
Stockouts / below reorder point / overstock	19 / 21 / 26 SKUs

1. Objective and Scope

An inventory audit tests whether the stock the system says exists is the stock actually on the shelf, and whether that stock is at the right level. This project answers four questions: how accurate are the records, where is the value lost, which items are out of stock or over-stocked, and which items deserve the most counting effort.

Deliverables

- Cleaned audit dataset (CSV) and the raw version it came from
- Discrepancy analysis by SKU, category and ABC class
- Stock-level classification using calculated reorder points
- Turnover and days-of-stock metrics, four charts, recommendations

Note on data

The data is synthetic. It contains realistic problems: shrinkage, unrecorded receipts, ghost stock (system shows units that are not there) and messy entries. Replace it with a stock export and a count sheet from a real warehouse to get real findings.

2. Dataset Description

Column	Meaning
SKU / Product / Category	Item identifier, name and group
Unit_Cost	Cost per unit in USD
Annual_Demand	Units sold or consumed in the last 12 months
Lead_Time_Days	Days from placing a supplier order to receiving it
Reorder_Point_Set	Reorder point currently configured in the system
System_Qty	Quantity the system says is on hand
Physical_Count	Quantity counted on the shelf during the audit

Sample rows (cleaned)

SKU	Category	Cost	Demand/yr	System	Physical	Variance
SKU-1125	Electronics	$140.78	159	16	16	+0
SKU-1052	Electronics	$195.34	817	142	130	-12
SKU-1075	Electronics	$66.22	518	84	84	+0
SKU-1016	Home & Kitchen	$55.45	814	448	448	+0
SKU-1008	Electronics	$105.08	718	9	9	+0
SKU-1074	Grocery	$5.98	378	40	40	+0
SKU-1058	Home & Kitchen	$20.96	274	0	0	+0
SKU-1079	Electronics	$167.83	700	44	44	+0

3. Data Cleaning

Four problems were found and fixed before any metric was calculated.

Issue	Count	Fix
Duplicate SKU rows	5	Kept the first row for each SKU
SKU codes with stray spaces or lower case	8	Stripped and upper-cased
Missing unit cost	4	Filled with the category median cost
Negative system quantity	3	Treated as a sign error and converted to positive

Result: 155 raw rows became 150 unique SKUs. Two of these fixes are assumptions. Filling cost with a median and flipping negative quantities are judgment calls; in a real audit each would be confirmed with the warehouse team rather than assumed.

In Excel: Data > Remove Duplicates on SKU; =UPPER(TRIM(A2)); and a median-by-category lookup for missing costs.

4. Analysis

4.1 Record accuracy and discrepancy

Metric	Value
SKUs audited	150
Exact matches	107 (71.3%)
Within 2% tolerance	107 (71.3%)
Discrepancy rate	28.7%
Shortages (physical below system)	-$46,678
Overages (physical above system)	$1,430
Net variance	-$45,248
Gross variance (absolute)	$48,108 (5.3% of system value)

Net variance can hide problems because shortages and overages offset each other. Gross variance is the better measure of how unreliable the records are.

4.2 ABC classification

SKUs are ranked by annual consumption value (annual demand x unit cost). Class A covers the first 80% of value, B the next 15%, C the last 5%.

Class	SKUs	% of SKUs	% of value	Inventory value	Exact-match accuracy
A	39	26%	80%	$751,839	67%
B	40	27%	15%	$69,367	72%
C	71	47%	5%	$39,143	73%

4.3 Variance by category

Category	SKUs	Net variance	Gross variance	Gross % of value	Accuracy
Electronics	18	-$34,666	$34,666	7.8%	78%
Apparel	26	-$6,075	$6,594	4.4%	58%
Home & Kitchen	21	-$3,503	$3,618	4.5%	67%
Stationery	39	-$9	$1,713	6.7%	72%
Tools	31	-$819	$1,311	0.9%	77%
Grocery	15	-$176	$206	2.0%	80%

4.4 Stock-level classification

Each SKU is tested against a calculated reorder point: ROP = daily demand x lead time + safety stock, where safety stock = 1.65 x 0.35 x daily demand x square root of lead time. The 1.65 gives about a 95% service level; the 0.35 demand-variability factor is an assumption. Overstock means more than 180 days of cover.

Status	SKUs	Share	Rule
Healthy	84	56%	At or above reorder point, up to 180 days of cover
Below reorder point	21	14%	Physical count below calculated ROP
Stockout	19	13%	Physical count is zero
Overstock	26	17%	More than 180 days of cover

4.5 Turnover

Annual cost of goods sold (annual demand x unit cost) is $4,106,829 against physical inventory of $860,348. Turnover = COGS / inventory value = **4.8x**, equal to about **76 days** of stock on hand. Because the data has no average-inventory history, year-end physical value is used as a proxy.

5. Key Insights

5.1 Largest discrepancies

SKU	Category	ABC	System	Physical	Value variance
SKU-1109	Electronics	A	626	510	-$18,029
SKU-1103	Electronics	A	626	519	-$12,074
SKU-1023	Apparel	A	126	90	-$2,791
SKU-1129	Apparel	A	1,273	1,219	-$2,765
SKU-1052	Electronics	A	142	130	-$2,344
SKU-1010	Electronics	A	65	28	-$2,220
SKU-1115	Home & Kitchen	A	67	42	-$1,242
SKU-1054	Home & Kitchen	A	139	117	-$1,141

5.2 Ghost stock

4 SKUs show stock in the system but have nothing on the shelf, worth $102 on paper. These are dangerous because the system will not trigger a reorder, so the shortage goes unnoticed until a customer order fails.

5.3 Critical stockouts

19 SKUs are at zero physical stock. Of these, 4 are Class A items: SKU-1029, SKU-1106, SKU-1121, SKU-1120. Class A stockouts carry the highest revenue risk and should be replenished first.

5.4 Overstock and tied-up cash

26 SKUs hold more than 180 days of cover. The amount above the target maximum level is worth about **$495,002**, which is cash that could be released by pausing reorders or discounting.

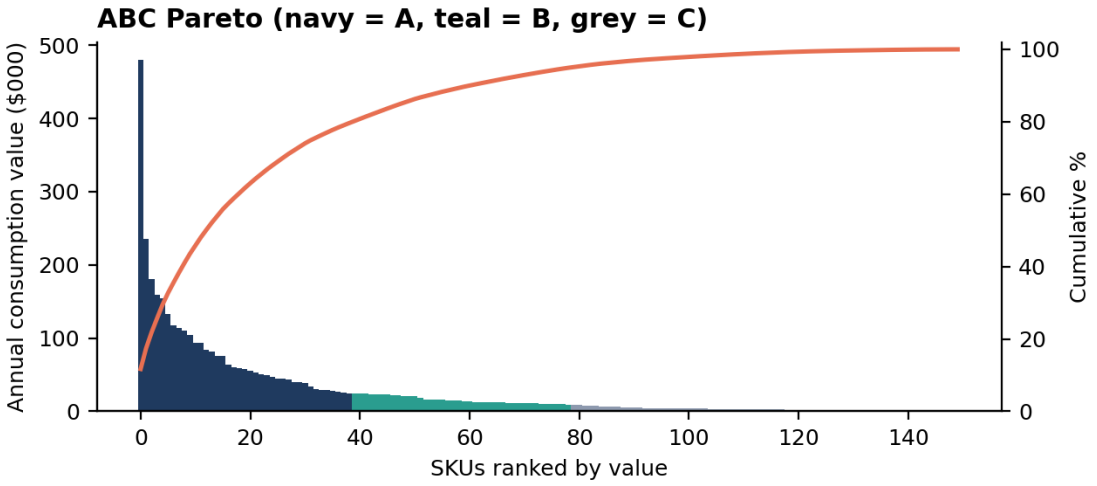
5.5 Reorder points in the system

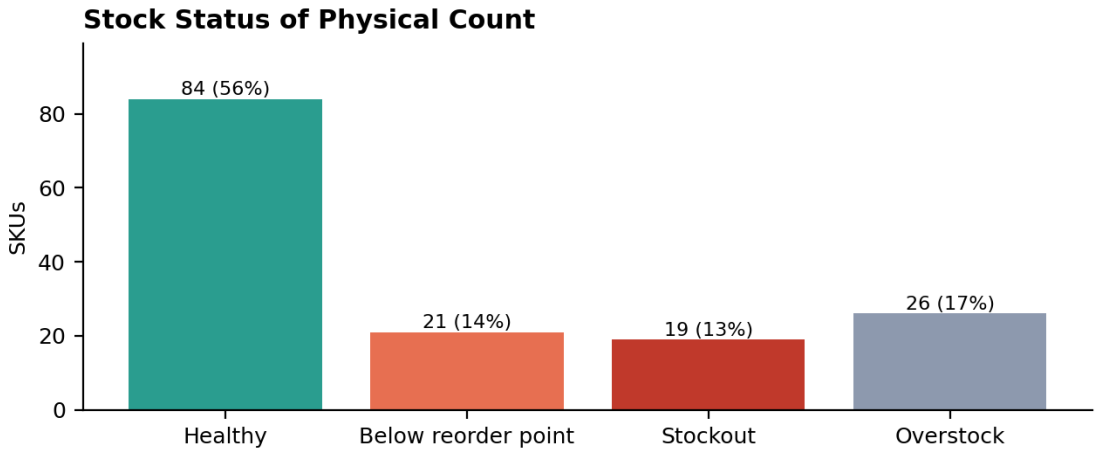
Comparing the configured reorder point with the calculated one: 34 SKUs are set more than 20% too low (risk of stockout) and 35 are set more than 20% too high (risk of overstock). Reorder points that were set once and never reviewed are a common cause of both problems.

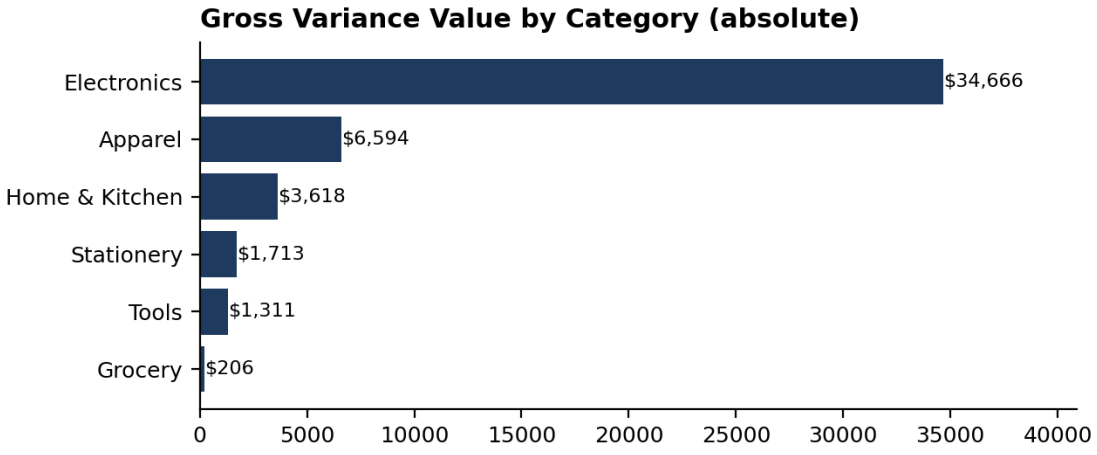
5.6 Accuracy by importance

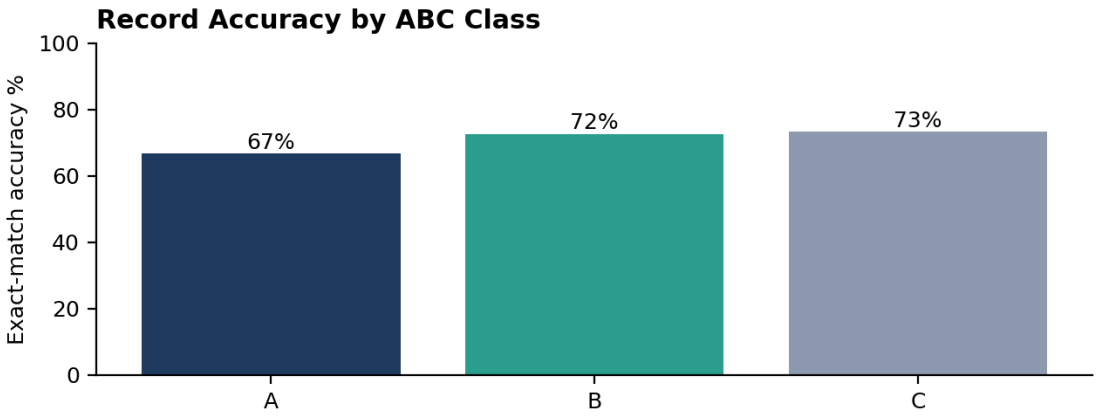
Class A accuracy is 67%, Class B is 72% and Class C is 73%. In a healthy operation the high-value items should be the most accurate, since they are counted most often.

6. Visualizations








Record Accuracy by ABC Class
Exact-match accuracy %
100
80
60
40
20
0
67%
72%
73%
A
B
C

7. Recommendations

1. Count by importance (cycle counting). Count Class A items monthly, Class B quarterly and Class C once a year. Class A holds 80% of consumption value in only 26% of SKUs, so this focuses effort where errors cost the most.

2. Investigate the worst category first. Electronics has the largest gross variance ($34,666). Review receiving, picking and returns processes there to find whether the cause is theft, damage or data-entry error. The data shows where the gap is, not why.

3. Adjust the system to the physical count, with approval. Post adjustments for the 43 mismatched SKUs only after root causes are checked, starting with the 4 ghost-stock items so reorders can trigger.

4. Reset reorder points. Replace the 69 materially misconfigured reorder points with calculated values and review them quarterly or when lead times change.

5. Release cash from overstock. Pause purchasing on the 26 overstocked SKUs. Releasing even half of the excess would free about $247,501.

Targets to track next audit: exact-match accuracy above 95% for Class A, a lower discrepancy rate, zero Class A stockouts, and turnover trending up.

8. Limitations

- Synthetic data; real counts also have counting errors, timing differences and in-transit stock.
- A single count date, so turnover uses year-end inventory as a proxy for the average.
- Demand variability (0.35) and the 95% service level are assumptions, and the 180-day overstock rule is a convention.
- Variance causes (theft, damage, mis-picks) cannot be determined from counts alone.

Appendix: Python script (core analysis)

```
import pandas as pd, numpy as np
d = pd.read_csv("inventory_audit_raw.csv")
d["SKU"] = d.SKU.str.strip().str.upper()
d = d.drop_duplicates(subset="SKU")
d["Unit_Cost"] = d.Unit_Cost.fillna(d.groupby("Category").Unit_Cost.transform("median"))
d["System_Qty"] = d.System_Qty.abs()

d["Var_Qty"] = d.Physical_Count - d.System_Qty
d["Var_Val"] = d.Var_Qty * d.Unit_Cost
accuracy  = (d.Var_Qty == 0).mean() * 100
gross_var = d.Var_Val.abs().sum()

d["Cons_Val"] = d.Annual_Demand * d.Unit_Cost
d = d.sort_values("Cons_Val", ascending=False)
cum = d.Cons_Val.cumsum() / d.Cons_Val.sum() * 100
d["ABC"] = np.where(cum <= 80, "A", np.where(cum <= 95, "B", "C"))

daily = d.Annual_Demand / 365
ss    = 1.65 * 0.35 * daily * np.sqrt(d.Lead_Time_Days)
d["ROP"] = (daily * d.Lead_Time_Days + ss).round()
d["Cover_Days"] = d.Physical_Count / daily

turnover = d.Cons_Val.sum() / (d.Physical_Count * d.Unit_Cost).sum()
```

Excel equivalents

- Variance: =Physical - System; value variance: =Variance * UnitCost
- Accuracy: =COUNTIF(VarianceRange,0) / COUNTA(SKURange)
- ABC: sort by consumption value descending, running total with =SUM($F$2:F2)/SUM($F$2:$F$151), then IF on 80% and 95%
- Reorder point: =Demand/365*LeadTime + 1.65*0.35*Demand/365*SQRT(LeadTime)
- Turnover: =SUM(ConsumptionValue) / SUMPRODUCT(Physical, UnitCost)